ON THE MOON

Francis Lyall*

I. Introduction

The news that potentially exploitable water may have been found in the deeper sunless craters of the north and south poles of the Moon brings closer the possibility of a return to our nearest neighbour. We will undoubtedly return to the Moon, but this detection, if confirmed, means we may well go back sooner than many of us had thought.

Water is necessary for the sustenance of life as we know it. In addition water can be split into hydrogen (fuel) and oxygen (life and fuel). If we can supply the water requirements of astronauts and other space-farers without having to lift it from the bottom of earth's gravity well, the exploration of the Moon (and beyond) will be so much easier. The return to the Moon, therefore, becomes the obvious venture to be undertaken next after the Space Station. Indeed, the possibility emerges that some states may choose to proceed with Moon projects without waiting for the Space Station to be completed. The news therefore raises once more the question of a fuller legal regime to deal with the many problems which the exploration and exploitation of the Moon may produce.

The Moon is romantic. It has always fascinated mankind.[1] Hanging in our sky as the deep cream of the full Harvest Moon or the faint silver crescent with its intimations of fullness to come or imminent decline, it is not to be wondered that tale-smiths sought it out. Stories of various 'Voyages to the Moon' start in the Seventeenth Century.[2] In our own century science fiction has used the Moon magnificently.[3] Now we have to deal with the reality.

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* Professor of Public Law, Faculty of Law, University of Aberdeen, Scotland, U.K.

1 *Cf.* M. Lachs, *Some Reflections on the State of the Law of Outer Space* 9 J. SPACE L. 3-12, at 3 (1981).

2 M.H. NICOLSON, VOYAGES TO THE MOON, (New York: Macmillan, 1948, 1960): P.B. GOVE, THE IMAGINARY VOYAGE IN PROSE FICTION, (New York: Columbia U.P., 1941; London: The Holland Press, 1961); L.T. SARGENT, BRITISH AND AMERICAN UTOPIAN LITERATURE, 1516-1875 (Boston: G.K. Hall, 1979).

3 B.W. ALDISS WITH D. WINGROVE, TRILLION YEAR SPREE: THE HISTORY OF SCIENCE FICTION, (London: Gollancz, 1986); *cf.* B.W. ALDISS, BILLION YEAR SPREE: THE HISTORY OF SCIENCE FICTION (London: Weidenfeld and Nicolson, 1973; Corgi, 1975). J. GUNN, THE ROAD TO SCIENCE FICTION, vol.1, *From Gilgamesh to Wells*, (New York: New American Library, Mentor Books, 1977). *See also* in J. CLUTE AND P. NICHOLS (Eds.), THE ENCYCLOPEDIA OF SCIENCE FICTION (London and New York: Orbit, 1993): *Critical and Historical Works about SF* (pp. 277-81) and *History of SF* (pp. 567-72).

Of course, the legal like the literary field is neither unknown, nor unploughed. Previous decades, and particularly the controversy surrounding the Moon Agreement of 1979, produced a considerable volume of discussion.[4] But the potential of imminent return to the Moon, and of developments thereon even short of full-scale exploitation, require that once more lawyers turn to the legal problems which the Moon may produce, and to their potential solutions. Too often the law has lagged behind the entrepreneur and the scientist.

For this, the twenty-fifth anniversary volume of the leading journal devoted to Space Law, I select only certain elements as we once more consider the question of the Moon, but in doing so draw attention to the fact that this Journal, even in its early years, was usefully concerned with such matters.[5]

II. International Law

One has, of course, to start with the fact that the legal arena is not unoccupied. The full title of the 1967 Outer Space Treaty indicates that its substance consists of 'Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies.'[6] Fundamentally, Art. III of the Treaty articulates the proposition that International Law applies in Outer Space. But most would go beyond that as a statement of treaty-law and assert that the proposition that International Law applies in Outer Space has passed into customary international law. This is important. If true, it means that a state, not being a member of the 1967 Treaty, must also recognise the role of International Law in space exploration, and the content of International Law for this purpose is not restricted to the generalities of the Outer Space Treaty.

Some areas may still hover on the cusp between treaty and custom, and therefore may or may not be binding. Other areas would seem to have a surer basis as law. Of them, of fundamental importance for this article is the principle that the Moon is not subject to national appropriation by claim of sovereignty, by use, occupation or any other means (Art. II, Outer Space Treaty). Fortunately the currently space-competent states are bound by their ratification of the 1967 Treaty. But I would hope that, were such a state to make use of the provision of Art. XVI as to withdrawal from the Treaty, it would continue to consider itself, and be held by appropriate

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4 *See,* for example, Bin Cheng, *The Moon Treaty: Agreement Governing the Activities of States on the Moon and other Celestial Bodies within the Solar System other than the Earth* 33 CURRENT LEGAL PROBS. 213-37 (1980), reprinted in his STUDIES IN INTERNATIONAL SPACE LAW 356-380 (Oxford: Clarendon Press, 1997); C.Q. Christol, *The 1979 Agreement Governing the Activities of States on the Moon and Other Celestial Bodies,* ch.7 of his THE MODERN INTERNATIONAL LAW OF OUTER SPACE 246-341 (New York: Pergamon Press, 1982).

5 *Cf.* S. Gorove, *Property Rights in Outer Space: Focus on the Proposed Moon Treaty,* 2 J. SPACE L. 27-30 1(974).

6 610 UNTS 205;(1968) UKTS 10,Cmnd.3519; 18 UST 2410, TIAS 6347.

process, to be bound as a matter of customary law by the rule otherwise articulated in Art. II.

Another approach to that question would be to call in aid the 1963 Declaration of Legal Principles Governing the Activities of States in the Exploration and use of Outer Space (GA Res. 1962 (XVIII)). Although there is debate as to the precise weight to be given to a UN Resolution in general as well as in particular instances,[7] the various Space Resolutions followed by compliant practice come closest to full legal competence, arguably being formative of customary International Law, albeit that the usual chronology of the creation process is inverted.[8] On this view Principle 3 of the 1963 Declaration, which anticipates the terms of the 1967 Treaty in setting aside outer space and celestial bodies from national appropriation, would continue to bind a state withdrawing from the 1967 Treaty. Such reasoning would apply also to the rule that activities in space and on the Moon have to be licensed and supervised by states (Art. VI, Outer Space Treaty; Principle 5, 1963 Declaration).[9]

Other questions, however, clearly remain treaty-based. Thus, that the Moon shall be used for peaceful purposes only, and that military bases are forbidden (Art. IV, para 2 Outer Space Treaty) are not matters of customary law, for military activities are not yet generally unlawful, and this proposition does not repeat within the UN Resolutions.

III. The Moon Agreement or another solution?

Few states have bound themselves to the terms of the one treaty whose major focus is the Moon, the Moon Agreement of 1979.[10] Arguments for and against the Moon Agreement are fairly stated in a report of the

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7 B. Sloane, *General Assembly Resolutions Revisited (Forty Years Later),* 58 BRIT.Y.B. INT'L L. 39-130 (1987); *cf.* M. Lachs, *The International Law of Outer Space* 3 R.C.A.D.I. 1-114, at 95-99 (1964); A.D. Terekhov, *UN General Assembly Resolutions and Outer Space Law,* 40 PROC. COLLOQ. L. OUTER SPACE 97-107 (1997)

8 I accept, of course, the Direct Broadcast Principles, 1982 G.A. Res. 37/92, as they were adopted by the vote of a majority which did not contain the major space-competent states.

9 There is room for further work on the duty and responsibilities of a state with regard to the activities of its nationals in this sphere: I think of the analogy of the legal status of privateers and of pirates. The one cannot be sanctioned; the other cannot be allowed to exist in space, *Moonraker* notwithstanding.

10 Agreement Governing the Activities of States on the Moon and Other Celestial Bodies, G.A. Res. 34/68, 18 INT'L LEGAL MAT'S. 1434 (1979). As at March 1998, the nine ratifying states were Australia, Austria, Chile, Mexico, Morocco, The Netherlands, Pakistan, The Philippines, and Uruguay. There are another five signatory states - France, Guatemala, India, Peru, Romania. As yet none in either category is independently space competent.

Congressional Research Service to a US Senate Committee in 1980.[11] The arguments there outlined would, of course, have different impacts in different countries, but it is useful to have them laid out in short compass.

The Moon Agreement was a well-intentioned effort to take the generalities of the 1967 Treaty further, and to spell out in more detail various matters with the Moon specifically in view.[12] It remains to be seen whether attempts will be made to resuscitate it, although at least one distinguished commentator considers this unlikely in the near future.[13] But whatever happens to the Moon Agreement, the fact is that it does identify several fundamentals. Its main problem is the 'common heritage' principle, with the implication of some 'Authority' as a hidden reef.

The 'Common Heritage' Principle

The 'common heritage' principle is, of course, a major reason why the Moon Agreement has not received more support. The statement that the Moon and its natural resources are the common heritage of mankind is baldly made in art. 11.1 of the Agreement. That paragraph goes on to say that the concept of 'common heritage' finds its expression in the rest of art. 11, and in particular in art. 11.5, which talks of an international regime and appropriate procedures.

The status and indeed content of the concept of 'common heritage' has been a matter of controversy.[14] But despite argument, the concept has come into use, and in particular has been sanctified by its application to The Area under the terms of art. 136 of the 1982 UN Convention on the Law

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11 Agreement Governing the Activities of States on the Moon and Other Celestial Bodies, Committee on Commerce, Science, and Transportation, U.S. Senate, 96th Cong., 2d Sess., Part 4, at 463-7. The whole Committee Report is valuable. Parts 1 and 2 were published in May, Part 3 in August and Part 4 in November 1980.

12 See n. 5 above.

13 C.Q. Christol, *The Moon Treaty and the Allocation of Resources*, 22(2) ANNALS AIR & SPACE L. 31-47, at 46-7 (1997).

14 S. Gorove, *The Concept of the "Common Heritage of Mankind"*, Ch. 5 of his STUDIES IN SPACE LAW 65-78 (Leiden: Sijthoff, 1977); (*cf. The Concept of the "Common Heritage of Mankind": A Political, Moral or Legal Innovation?* 9 SAN DIEGO L. REV. 390 (1972). G. M. Danilenko, *The Concept of the "Common Heritage of Mankind" in International Law*, 13 ANNALS AIR & SPACE L. 247-63 (1988); H. S. Rana, *The "Common Heritage of Mankind" and the Final Frontier*, 26 RUTGERS L. J. 225 (1994).

of the Sea.[15] While it is true that the concept is not yet a general rule of International Law, in the Law of the Sea it has found a working role.

Be that as it may, it is coming to be accepted that, in any manifestation of it, the 'common heritage' concept has particular elements: that certain regions should not be subject to national appropriation in any way, that there will be a management system for such an area, that the managers, be they state or international organisation, will act as representative of mankind, that any benefits from such areas will be shared internationally, and that the area will be used for peaceful purposes only.[16] Correlatively, scientific research in the area is encouraged. Ideally this would apply to the Moon.

Although the idea of 'common heritage' first emerged in relation to Outer Space, its main development has been in the Law of the Sea. While it is true that there have been statements separating the notion as it is found in the Law of the Sea and in Space, as time goes on it is not likely that separation can usefully be maintained. In the meantime, however, the history of the concept in the Law of the Sea shows clearly the difficulties that some states have in accepting it. In brief, states which have the technology to undertake activities which may be hindered or restricted by a 'common heritage' regime, were reluctant to sign up to such a system. Indeed, the modification of Part XI of the 1982 Convention has yet to be proved to work satisfactorily, although that modification has been sufficient to induce the US and the UK to move towards ratification. The same pattern might prove to be true of the Moon Agreement. Space-competent states have been reluctant to sign, let alone ratify, the Moon Agreement because of art. 11. However, in time, things may change.

One element of debate is that the 'common heritage' idea conflates law and morality, and this is said to be unacceptable. Curiously the argument seems to be put most strongly by those anxious to defend limited commercial interests.

Morality and Law

To put the matter contentiously: the 'common heritage' principle can be attacked on the ground that it is an attempt by states unable themselves to exploit resources because of technical or other incompetence, to compel space competent states to invest time, trouble and finance in a project, and then divert to non-participants and non-investors some of the rewards of these entrepreneurial activities. On this view the 'have-not' nations are free-loading. There may be some force in this argument. Certainly it will be interesting to see whether in the Law of the Sea manifestation of the concept, the benefits of that 'common heritage' do in fact make their way

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15 The Law of the Sea Convention, Montego Bay, Jamaica, 1982, 21 INT'L LEGAL MAT'S 1261-1354 (1982); 1983 UKTS Misc. No. 11, Cmnd. 8941; G.A. Res. 48/263 amending Part XI of the Convention is printed at 33 INT'L LEGAL MAT'S 1309 (1994) and President Clinton's Letter to the US Senate of Transmittal of the Convention with recommendation of approval is at 34 INT'L LEGAL MAT'S 1393-447 (1995).

16 *Cf.* C.C. Joyner, *Legal Implications of the Concept of the Common Heritage of Mankind*, 35 INT'L & COMP. L. Q. 190-99 (1986); Christol, n. 13 above.

down to the welfare of the peoples of the less-developed world and does not simply augment the extra-territorially held coffers of their rulers.

But, although there is opportunity for abuse, that is not a sufficient reason to reject the concept of 'common heritage'. The separation of morality and law which often underlies the arguments of those who reject the concept, is pernicious. It is corrosive of the integrity of a legal system. Legal systems do not continue long to work when those whom they affect concentrate on the letter and ignore the spirit, when legal propositions are construed as mere formal requirements, when the law is seen as part of the game to be used to defend advantage and impede the efforts of the other side, or when men seek to adapt or amend agreements for their own selfish advantage. Unfortunately we have seen examples of this in Space Law, particularly within the deliberations of the International Telecommunication Union, but it is also evident in the debates of the international trade organisations.[17]

The fact is that Space Law does already contain statements as to the 'benefit' of mankind.[18] 'Common heritage' is another of these concepts, and it is grievous error simply to shoot it down in flames. Its core does represent a moral imperative, which should be preserved and fostered.[19] If it is not, it is not too much to say that barbarity beckons only a few decades hence. 'Have' and 'have-not' are too intertwined nowadays for the one to neglect the interests of the other. As Clausewitz pointed out, war is the continuation of policy by other means - and modern war need not be clinical if the combatants are desperate or irrationally inflamed.

Another Agreement?

Some form of general agreement on Moon matters is highly desirable. Ideally this should be agreement on a global scale, with a general acceptance similar to that of the Outer Space Treaty. An obvious model is the 1982 Law of the Sea Convention, which, in its Part XI, as amended, sets up the International Sea-Bed Authority to supervise and regulate developments in the Area beyond national jurisdiction, the 'common heritage of mankind' designated by art. 136 of the Convention.

Whether a new Moon Agreement should set up a 'lunar authority' to supervise, and perhaps to license and draw a revenue from exploitative activities is a neat question. Utopians would prefer the model of the International Sea-Bed Authority, together with the concept of 'common

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17 I hope to write more generally on this point elsewhere.

18 Cf. Art. I of the Outer Space Treaty, and the Declaration of International Cooperation in the Exploration and Use of Outer Space for the Benefit and in the Interest of All States, Taking into Particular Account the Needs of Developing Countries, G.A. Res. 51/122 (1996).

19 *Cf.* C.Q. Christol, *Outer Space and Exploitability: International law and developing nations,* 11 SPACE POL'Y 146-60 (1990).

heritage' to ensure that benefits are spread widely. Such, however, seems unlikely.[20]

More practically perhaps, an agreement between the states active in space exploration, could suffice.[21] The 'common heritage' wording could be avoided within such an agreement should it remain unduly awkward for major participants. This points to a model which has worked pretty well for the last almost four decades.

At present the more flexible, non-authoritarian Antarctic system seems to me the model that should be adopted. Based on the Antarctic Treaty,[22] the system developed by the Parties to the Treaty has the merit of having been developed slowly and in response to practical needs, for precisely the sort of scientific enterprise that is likely at first at least to be seen on the Moon. If there is indeed recoverable water resources frozen in the bottom of deep craters near the Moon's poles, there may be rush to set up bases there. The Antarctic model would allow for that. Later, if significant resources are discovered in some other particular and restricted areas of the Moon, something legally stronger and more structured will be required.[23] Such a development would logically at least require some sort of regime to regulate competition, and give a good 'title' to those involved.[24] The history of the Antarctic regime provides such a model.[25] Of course, in the further future one might hope for its replacement by a fuller structure on the Sea-Bed Authority model, but we need not hurry to that: a basic agreement is better than nothing.

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20 *Cf.* Christol, n. 19.

21 This could be eased if Moon exploration and exploitation were a joint effort. The European Space Agency, for example, is already a consortium.

22 The Antarctic Treaty, 1959; 402 UNTS 71; (1961) UKTS 97, Cmnd. 1535; 12 UST 794, TIAS No. 4780.

23 See D.H. Overholt, *Environmental Protection in the Antarctic: Past, Present and Future*, 28 CAN. Y. B. INT'L L. 227-62 (1990); C. Redgewell, *Current Developments: International Law, III Antarctica*, 39 INT'L & COMP. L. Q. 474-81 (1990).

24 Note that the word 'title' is here in inverted commas. The most that is likely is a lease or license for a limited time and area: cf. art. 137.2 of the Law of the Sea Convention, 1982 which restricts the nature of legal titles involved in the Area and to minerals and other resources extracted. Although by art. 11 3 of the Moon Agreement property rights on the Moon are excluded, by art. 6.2 states members of that Agreement have a right to use appropriate quantities of mineral and other substances found on the Moon for the support of their missions, as well as to collect and remove samples.

25 *Cf.* C.G. Joyner, *The Antarctic Treaty System and the Law of the Sea - Competing Regimes in the Southern Ocean?*, 10 J. MARINE & COASTAL L. 301-311 (1995). *See also: Antarctic Treaty Consultative Parties: Measures relating to the Furtherance of the Principles and Objectives of the Antarctic Treaty [Meetings I-XIX]*, 35 INT'L LEGAL MAT'S 1165-77 (1996); and, *Idem:, Recommendation XVIII-1 on Tourism and Non-governmental Activities*, 35 INT'L LEGAL MAT'S 1178-86 (1996).

IV. SAHA: a Moon Crater as a Science Preserve

Whatever the outcome of discussions (or inertia) as to the formal legal position affecting the generality of the Moon, there is one matter that needs to be dealt with urgently. Certain areas of the Moon should be set aside for scientific study and endeavour. In broad, no-one would quarrel with such a proposal, and art. 9 of the Moon Agreement might be used to provide a legal base for it. But, whatever happens, one particular scheme deserves special consideration, care and protection, whether we have a whole-lunar agreement, or a less formal arrangement. This is the proposal to set aside one or more deep craters on the lunar far-side for the activities of radio-astronomers, including those who listen in the hope of detecting evidence of extra-terrestrial intelligence.[26]

Radio-astronomy makes a passive use of the radio-spectrum. The radio signals radio-astronomy deals with are not generated by the instrumentation used. Radio emissions from far into space are detected, recorded and analysed, and many important scientific discoveries have been made. Radio-astronomy is therefore an important human activity.

However, the protection given to radio-astronomy within the Radio Regulations of the International Telecommunication Union, is not satisfactory. Freedom of interference from man-made radio emissions in those sections of the radio spectrum which are most useful for the science is not adequately guaranteed. Not all important wave-lengths within the usable spectrum are set aside for passive use. Allocations within the Radio Regulations are now extending into bands previously unused, and sections of the spectrum relevant for radio-astronomy are now being subjected to encroachment, particularly by the new satellite telephony systems. And all this is entirely apart from the problems which can be caused by radio emitters which are not efficiently tuned, properly screened, damped or otherwise neutralised.[27]

Many of the problems of radio astronomy in general, and of SETI in particular, would be alleviated were radio-telescopes to be fully shielded from man-made radio interference. The proposal therefore has been made that an appropriate site would be within a crater on the far-side of the

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26 The Search for Extra-Terrestrial Intelligence (SETI), is a curious, but potentially very important sub-set of the use of radio-telescopes.

27 For example, a badly screened micro-wave oven can produce a radio signal which could interfere with or mask a spectrum band being used by a radio-telescope.

Moon, the crater Saha being particularly identified as suitable the main proponent of the idea having outlined his questions in 1994.[28]

By Art. I of the Outer Space Treaty, outer space including the Moon and celestial bodies is free for exploration and use by all, and this use specifically includes scientific investigation. As indicated above, by Art. VI of the Outer Space Treaty what is done in space or on the Moon is required to be subject to the supervision of the appropriate national state having jurisdiction over those conducting the relevant activity. Other general aspects of International Law indicated above, including the question of 'nuisance', would also have relevance. In this discharge of their duty of supervision, states ought to ensure that the activities of nationals do not impede the activities of others.[29]

But that is obviously insufficient protection for the radio-astronomers' projects. What is required is a proper formal setting aside of a specific site, and its protection for radio astronomic purposes, which we expect would include SETI. This should be done before any further steps are taken in Moon exploration.

The limited terms of the Moon Agreement as it stands at present do not provide sufficient protection for a far-side site on the Moon for radio-astronomy purposes. While its terms make lawful landing and the placing of facilities, stations and installations anywhere on the Moon (art. 8.1-2 and 9), art. 8.3 merely requires that activities of states parties to the Agreement shall not interfere the activities of other states parties on the Moon. This is quite vague and imprecise. This 'interference' is not a reference to 'radio interference' though arguably the term could encompass it. But the argument is not strong: any competent lawyer could blow holes in it, deny the duty in a particular instance, or justify an alleged breach by his state. Article 8.3 of the Moon Agreement is therefore insufficient for the protection of a Saha project.

On the other hand, were the Moon Agreement to be revived and gain a general membership particularly among the space-faring nations, its terms could be used. Once exploitation is about to become feasible, parties to that treaty are to agree on a regime for the exploitation and use of the Moon (art. 11.5). Setting aside a Saha site could be accomplished within that regime, whether the regime comprises a formal structure on the lines

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28 J. Heidmann, *Saha Crater: A Candidate for a SETI Lunar Base*, 32 ACTA ASTRONAUTICA 471-2 (1994); *What legal questions are raised by the establishment of a dedicated lunar farside specific crater for high sensitivity radioastronomy?*, 37 PROC. COLLOQ. L. OUTER SPACE 255 (1994), (*cf.* S. Doyle's comments on Dr Heidmann's proposal, summarised 37 PROC. COLLOQ. L. OUTER SPACE 302 (1994); *Recent progress on the Lunar Farside Crater Saha Proposal*, a paper presented to the 48th International Astronautical Congress, Turin, Italy, 1997: IAA-97-IAA.9.1.05; *Cf.* A.A. Cocca, *Reservation of a Lunar Zone for SETI Purposes*, 38 PROC. COLLOQ. L. OUTER SPACE 270-3 (1995).

29 This could cover both the technical question of 'interference' as that word in known within the ITU system, and also its broader meaning within International Law: cf. the line of thought stemming from the Trail Smelter Arbitration (1938-41), 3 R.I.A.A. 1905.

of the International Sea-Bed Authority, or the less bureaucratic regime that applies in the Antarctic.

But, irrespective of which of these alternatives comes to be, some sort of regime will be required to provide ground rules at least between participants in the exploration and exploitation of the Moon. However such matters are arranged, care should be taken to ensure that the fullest protection is given to the requirements of a far-side lunar radio observatory under any Moon regime. Thereafter, once the legal basis has been established for the site, the ordinary procedures of assignment and notification would be gone through for the site's use of telecommunications services. Its protection from interference would, however, be dependent on the enforcement of the appropriate provisions of the Radio Regulations.

V. Conclusion

Twenty-five years on from the first issues of this Journal, the problem of the Moon has been brought once more to the attention of lawyers. The prospect is that the Moon may soon be revisited, on a more permanent basis than hitherto. It would be easy to await developments, and let 'things work themselves out'. It would be more prudent deliberately to choose the route of future developments. An 'Antarctic solution' may be easier to agree on than a 'lunar authority' drawn up on the model of the Law of the Sea. But whichever avenue is taken, steps should be taken now to secure the interests of radio-astronomers.